Dr Bhushan Atote

07442 575125 ◇ 309H, Millennium View, Coventry CV1 1JB
Bhushan.Atote@warwick.ac.uk ◇ https://www.linkedin.com/in/bhushanatote/

EDUCATION & QUALIFICATIONS

PhD in Computer Science March 2022 – Aug 2026
University of Warwick, Coventry, UK
Thesis: *Explainable AI for Discriminative and Generative Vision Models*

Master of Engineering in Computer Engineering 7.98/10 Aug 2014 – Nov 2016
Maharashtra Institute of Technology, SPPU, Pune, India
Thesis: *Optimization of signal behavior through Dynamic Traffic Control*

Bachelor of Engineering in Computer Engineering Aug 2009 – Aug 2013
RTMNU, Nagpur, India
Major: *Computer Science*, Minor: *Mathematics*

WORK & LEADERSHIP EXPERIENCE

Senior Graduate Teaching Assistant Oct 2022 – Feb 2026
The University of Warwick, Department of Computer Science
Courses: *CS413(Image and Video Analysis), CS342(Machine learning), CS910/430(Foundations of Data Analytics), CS355(Digital Forensics) and CS139(Web Development Technologies)*
The University of Warwick, Warwick Business School
Courses: *IB9YC0(Data Driven Decision Making), IB9JB0(Marketing & Strategy Analytics)*
The University of Warwick, Warwick Summer School
Courses: *Data Science and Machine Learning: The Fundamentals*

- Evaluated student assignments, and projects, providing timely and constructive feedback
- Analysed and resolved student queries related to course material during contact hours and lab sessions
- Maintained accurate records of student attendance and participation
- Graded and provided feedback on coursework, ensuring consistency and fairness in evaluation process
- Supervised and guided multiple projects spanning medical imaging, digital forensics, and marketing analytics.

PGR Teacher Champion May 2023– Dec 2023
The University of Warwick

- Collaborating with the PGR lead, Academic Development Centre, and previous WPTC members to design and implement new components of the program that will support PGR teachers
- Ensuring the continuation of essential WPTC resources, such as the Journal of PGR Pedagogic Practice (successfully produced issue 3 on EDI), and the WPTC Moodle resources
- Collaborating with fellow PGRs within the WPTC group using Microsoft Teams to coordinate efforts and share ideas
- Participating in regular meetings with the WPTC team, staying informed about the program's progress, and communicating effectively and promptly with team members

Academic Services Assistant May 2023 – June 2023
The University of Warwick, Warwick Business School

- Coordinating and supporting examination activities, including implementing arrangements for special requirements, proofreading examination papers, and invigilation for both in-person and online exams
- Handling academic integrity issues, including drafting communications related to misconduct cases and promoting good academic practices in student meetings
- Maintaining office systems and records, including databases related to academic conduct and examinations, and updating Management Information Systems
- Providing general administrative support, including meeting arrangements, note-taking, and stationery orders, and performing other duties as directed by the Administrative Officers (Academic Services)

Research Assistant May 2022 – Sept 2022
The University of Warwick, Department of Computer Science
Project title: *Monitoring and recognising drivers' activities for cars*

- Conducted literature review of state of the art models
- Managed in data preparation and analysis

Associate Consultant April 2018 – Oct 2020
CapGemini Technology Services India Limited, India

Project title: *Whales - Recognize Me!*

- Understood the research problem and prepared multimodal dataset combining hydrophone acoustic recordings and sperm whale fluke images for individual profiling
- Designed model architecture integrating CNN-based fluke re-identification with acoustic coda/click pattern analysis, and formulated validation strategy using confirmed sighting data
- Applied transfer learning with pre-trained CNN backbones, fine-tuned via Siamese/triplet-loss setup, and tuned hyperparameters to optimize identity-matching accuracy
- Carried out data augmentation and pre-processing (denoising, click detection, coda segmentation) to improve model robustness, and built similarity-based matching to fuse acoustic and visual signals into unified whale profiles
- Validated model outputs against known individual whale catalogs in collaboration with domain researchers, iterating on newly collected field data to improve performance

Project title: *Sentiment Analysis on Travel Portal*

- Analysed business problems using statistical models and data-driven techniques to generate actionable insights and develop predictive/prescriptive analytics solutions.
- Developed NLP models for topic extraction, sentiment analysis, and text classification of unstructured client data, achieving a 0.957 validation accuracy.
- Optimized classification models to minimize false-positive rates and categorized customer feedback into positive and negative sentiment clusters.
- Designed analytical reports and compared model performance with RNN/LSTM approaches to evaluate and improve predictive accuracy.

Graduate Teaching Assistant Aug 2014 – July 2016
Maharashtra Institute of Technology, SPPU, Pune, India

- Created and maintained lab manuals for Python and Machine Learning courses
- Delivered presentations on various Machine Learning topics to graduate students
- Mentored and resolved queries of graduate students during lab sessions
- Supervised a winning team of 5 undergraduates in Smart city project intercollegiate competition

PROFESSIONAL SERVICE

Editor:
Journal of PGR Pedagogic Practice, Issue 3 2023
Peer Reviewer:
IEEE Signal Processing Letters, ICML, ECCV 2024, 2025, 2026
Programme Committee Member:
AI for Multimedia Forensics & Disinformation Detection Workshop, ECCV 2026 2026

ADDITIONAL EXPERIENCE

SSLC, Secretary April 2022 – Dec 2022
University of Warwick, Department of Computer Science

- Prepared and circulated SSLC meeting minutes to committee members for approval and maintained the SSLC webpage with updated documentation.
- Coordinated SSLC meetings and PGR feedback sessions, including scheduling around committee availability, communicating arrangements, and supporting GTAs in raising and addressing relevant issues with staff.

WIHEA Fellow Oct 2023 – Sept 2024
University of Warwick

- Participated in three Learning Circles - AI in Education, Internationalisation, and Developing Academic Literacies, as part of the Fellowship, with a particular focus on applying AI research to enhance teaching and learning practices at the University of Warwick.
- Gained broader academic perspectives through interdisciplinary engagement, exploring diverse educational strategies and approaches to learning.

SKILLS & INTERESTS

- **Languages:** Python, MATLAB, R, HTML/ CSS, JavaScript
- **Frameworks:** PyTorch, HuggingFace, OpenCV, Tensorflow, NLTK, Sklearn, NumPy, Pandas, Matplotlib

- **Tools and Technologies:** MLflow, MySQL, LaTeX, Git, Linux, Stata, Tableau, PowerBI

PUBLICATIONS

Refereed Conference Papers

- **B. Atote**, V. Sanchez: *InterDDPM: Interpretable Diffusion Denoising Probabilistic Model for image understanding with the help of human-comprehensible prototypes.* ***IEEE TIP*** *(Under review)*
- **B. Atote**, V. Sanchez: *How were you created? Explaining synthetic face images generated by diffusion models*, Computer Vision – ECCV 2024 Workshops. **ECCV 2024**. Lecture Notes in Computer Science, vol 15644. Springer, Cham. DOI: 10.1007/978-3-031-92089-9_17
- **B. Atote**, V. Sanchez: *Enhanced Prototypical Part Network (EPPNet) for Explainable Image Classification via Prototypes*. **IEEE**, to be published in International Conference on Image Processing (**ICIP**), Abu Dhabi, UAE, 2024. DOI: 10.1109/ICIP51287.2024.10647799
- Y. Ma, V. Sanchez, S. Nikan, D. Upadhyay, **B. Atote** and T. Guha: *Robust Multiview Multimodal Driver Monitoring System Using Masked Multi-Head Self-Attention*. **IEEE/CVF**, Conference on Computer Vision and Pattern Recognition Workshops (**CVPRW**), Vancouver, BC, Canada, 2023, pp. 2617-2625, DOI: 10.1109/CVPRW59228.2023.00260
- **Bhushan Atote**, Saniya Zahoor, Taranpreet Singh Saini, Mangesh Bedekar: *Optimization of Signal Behavior through Dynamic Traffic Control: Proposed Algorithm with Traffic Profiling*. **IEEE**, 2nd International Conference on Contemporary Computing and Informatics, Amity University, Noida, India; 12/2016, DOI:10.1109/IC3I.2016.7918033
- **Bhushan Atote**, Saniya Zahoor, Taranpreet Singh Saini, Mangesh Bedekar: *Time Alone is Not the Clear Indicator of Relevancy*. **IEEE**, 2nd International Conference on Contemporary Computing and Informatics, Amity University, Noida, India; 12/2016, DOI:10.1109/IC3I.2016.7918041
- **Bhushan Atote**, Mangesh Bedekar, Saniya Zahoor, Suja Panicker: *Proposed Used of Information Dispersal Algorithm in User Profiling*. **SPRINGER**, ICT4SD, International Conference on ICT for Sustainable Development, Goa, India; 07/2016, DOI:10.1007/978-981-10-3932-4_9
- **Bhushan Atote**, Mangesh Bedekar, Suja S. Panicker: *Centralized Approach towards Intelligent Traffic Signal Control*, **ACM**, ICTCS-2016, Second International Conference on ICT for Competitive Strategies, Udaipur, INDIA; 03/2016, DOI:10.1145/2905055.2905121
- **Bhushan Atote**, Saniya Zahoor, Taranpreet Singh Saini, Mangesh Bedekar: *Inferring Emotional State of a User by User Profiling*. 2nd International Conference on Contemporary Computing and Informatics, Amity University, Noida, India; 12/2016, DOI:10.1109/IC3I.2016.7918021
- **Bhushan Atote**, Mangesh Bedekar, Saniya Zahoor, Bharat Dangra: *Personalization in User Profiling: Privacy and Security Issues*. **IEEE**, International Conference on Internet of Things and Applications (IOTA-2016), Maharashtra Institute of Technology, Pune, India; 01/2016, DOI:10.1109/IOTA.2016.7562763

AWARDS

- **WIHEA Fellow** Oct 2023 - July 2024
  Contributed my expertise to projects that incorporate cutting-edge technology in higher education pedagogy
- **National Overseas Scholarship** March 2022
  Funding provided for PhD by Government of India
- **Graduate Aptitude Test in Engineering(GATE) Qualified** Feb 2014
  Secured financial support in the form of a monthly stipend from AICTE/MHRD (Indian government agencies) during Master of Engineering studies
- **Chess Championship** Aug 2010
  Secured Runner-up position in Undergraduate Chess Championship